Determining the frequency of transmitted waves with a Lloyd's mirror interferometer

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Transmitted waves can either add together in constructive interference or effectively cancel one another out in destructive interference. A Lloyd's mirror interferometer is a setup of an electromagnetic wave transmitter, a mirror, and a receiver that produces a measurable interference pattern based on the distance each component is place from one another. Finding the parameters of the setup that lead to constructive interference and a using a least squares regression relationship, the frequency of the transmitted wave was calculated. The frequency of the transmitted wave produced by the Pasco microwave transmitter is 10.5 plus or minus 0.1 GHz.

I. INTRODUCTION

Humanity has used electromagnetic waves to communicate for more than a century. Radio waves were first employed by Guglielmo Marconi for long-distance communication in 1895 and other forms have been used all across the world since that time [1]. However, the interaction of these waves goes beyond mere transmitting and receiving; when two or more electromagnetic waves converge, they can constructively and destructively interfere.

Constructive interference occurs when waves of the same frequency align their crests and troughs. The amplitudes of the waves sum together and the resultant wave is a more intense wave than any of the individual waves. Conversely, if any two or more waves have crests aligned with troughs and vice versa their amplitudes will be subtracted from one another and the resultant intensity at that location is less than the intensity of any of the individual waves [2].

Interferometers are devices used to test the constructive and destructive interference patterns of electromagnetic waves. Named after its inventor Humphrey Lloyd, a Lloyd's mirror interferometer uses a transmitter and receiver that have large output ranges. Since the waves propagate from the transmitter across a wide range, portions of the wave that are not directed towards the receiver can be redirected towards the receiver by a reflective surface. The two beams are then recombined at the receiver, leading to interference patterns that result from the phase difference between them, or distance between peaks in the waves [3].

II. KEY FINDINGS

A. Experiment Setup

The setup includes a Pasco microwave transmitter which outputs an electromagnetic wave and a Pasco microwave receiver to receive this wave, both of which have rectangular cone-shaped horns. The transmitter produces a beam coming out of its horn, while the receiver only detects radiation coming in its horn. Because of the way electromagnetic waves reflect off the walls of the horn, the effective center of the transmitter and the effective point where the receiver receives is five centimeters inside the open end of the horn.

The experiment setup is illustrated in fig. 1. The reflector is an aluminum sheet placed midway between the transmitter and receiver, with the flat face parallel to the direction of transmission, and slightly out of the path of the direct line between the transmitter and receiver. Portions of the wave will travel directly from the transmitter to the receiver and other portions of the electromagnetic wave will be reflected off the aluminium sheet redirecting them towards the receiver. The distance at which this sheet is placed measured perpendicular to the straight line connecting the transmitter and receiver will

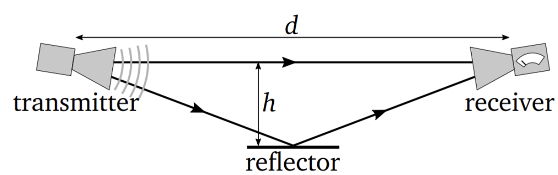

FIG. 1. Diagram of the experiment setup. The distance d is the distance between the effective transmission and reception points on the transmitter and receiver. The distance h is the distance from the directly transmitted wave to the reflector perpendicular to the direct wave motion.

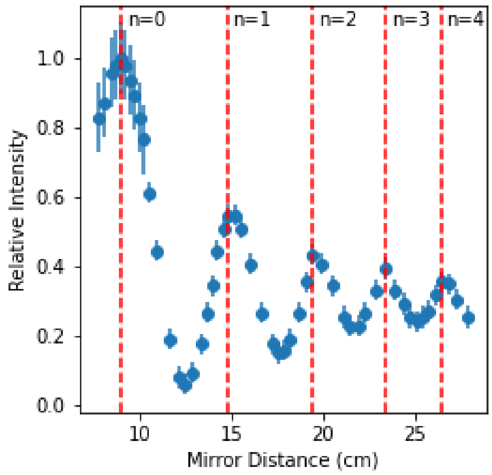

FIG. 2. Plot of the measured relative intensity of the transmitted wave against the mirror distance with local maximums identified. Each maximum is a constructive interference distance and has been labeled sequentially with an integer number. The differences in the uncertainty between various measurements is associated with the amplification factor of the receiver. The lower values of h had a higher uncertainty due to a greater amplification.

determine whether or not the waves will constructively or destructively interfere.

A two meter long ruler was taped to a table. This was used as a straight edge and a measurement device to position the effective point of transmission and reception for the transmitter and receiver facing one another 1.029 ± 0.001 m apart. The same ruler was also used to position the horizontal location of the reflector. It was placed at the halfway point between the transmitter and receiver and the sheet is wide enough to include the true halfway point. We measured the distance between the reflector and the line of direct transmission between the transmitter and receiver as well as the intensity displayed on the dial of the receiver for increments of approximately four millimeters in change to the reflector distance ranging from 7.7 to 27.9 centimeters. This change in distance was determined so as to include several data points between each peak and valley to ensure that all constructive and destructive interference peaks were observed.

The distances h that led to a maximum in relative intensity were determined to be positions in which the directly transmitted wave and the reflected waves were constructively interfering. The relative intensity is a value that is related to the intensity of the wave, various amplification factors were used on the receiver depending on the intensity of the wave so the relative intensity is a normalized method of comparing these data points. Using this information, one can deduce the wavelength and frequency of the wave being transmitted.

B. Mathematical Background

Constructive interference occurs between two waves of the same frequency when the peaks line up with one another. If they are transmitted by the same source, this is equivalent to the statement that the difference in the distance the waves have traveled is an integer multiple of the wavelength of the wave. For a direct path length L and a reflected path length L' for a wave of wavelength λ, constructive interference occurs when

$$L' - L = n\lambda, \tag{1}$$

for every $n \in \mathbb{Z}$.

In the setup outlined in fig. 1 the directly transmitted path length L is

$$L = d, \tag{2}$$

the distance between the transmitter and receiver. Recalling the Pythagorean theorem, the reflected path length L' is

$$L' = 2\sqrt{\frac{d^2}{4} + h^2}. \tag{3}$$

Combining eq. (1), eq. (2), and eq. (3) yields the following relation

$$2\sqrt{\frac{d^2}{4} + h^2} - d = n\lambda. \tag{4}$$

We measure the relative intensity as a function of the mirror distance h to find the distances leading to constructive interference. Each of these peaks will be assigned a sequential integer number and the difference in path length will be calculated. The slope of the line between the path difference and the integer numbers associated with multiples of the wavelength will yield the wavelength of the wave and from there the frequency can be derived. This method operates under the Nyquist theorem which is the assumption that the incremental distances we move the mirror by is small enough to observe every constructive interference peak.

The frequency ν of the wave is related to the speed of light c by

$$\nu = \frac{c}{\lambda}. \tag{5}$$

C. Results

The plot of the intensity measured by the receiver as a function of the mirror distance is shown in fig. 2. The peaks in the plot are indicative of constructive interference between the directly transmitted and reflected waves which add together to produce a more intense wave. The local minima are destructive interference which are associated with a lower intensity wave. The mirror distances

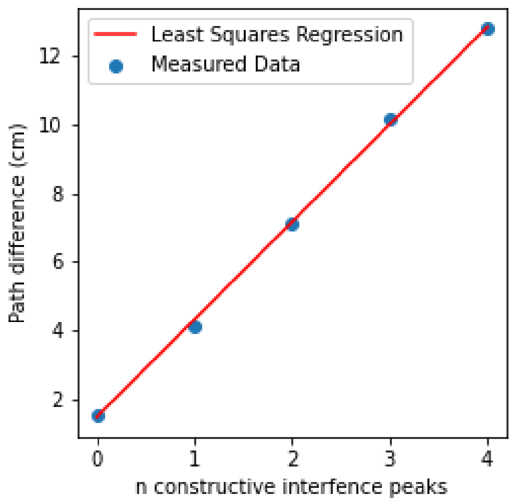

FIG. 3. Plot of the path difference between the directly transmitted and reflected waves at measured constructive interference locations. A linear least squares regression line is plotted, the slope of which is the wavelength of the transmitted wave.

that yield constructive interference were sequentially labeled with integer numbers. The mirror distances are the h values and the integers are the n values in eq. (4).

The path difference between the directly transmitted and reflected waves at the identified constructive interference mirror distances are plotted in fig. 3 as a function of the integer value associated with each distance. The theory indicates that the slope of the line between each constructive interference location and the path difference is the wavelength of the transmitted wave. The actual integer values are not indicative of anything, they just need to be sequential as they are counting the wavelengths of the wave.

Using the known value for the speed of light (3×10^8 m/s) and the slope from the least-squares regression line in fig. 3, we have determined that the frequency ν of the transmitted wave is

$$\nu = 10.5 \pm 0.1 \text{ GHz}. \tag{6}$$

The uncertainty on this measurement is incredibly small (approximately .1%) because of the fine adjustments of mirror distance and double measurements taken of all length measurements to ensure accuracy.

III. CONCLUSION

An interferometer consisting of an electromagnetic wave transmitter, receiver, and an adjacent mirror of varying distances produces an interference pattern that can be used to calculate the frequency of the transmitted wave. The Pasco microwave transmitter used in this experiment was found to transmit a wave with a frequency of 10.5 ± 0.1 GHz.

The interferometer could have been set up in a fashion where the speed of light was measured instead of the frequency. Future work will include using the Pasco microwave transmitters to measure other physical quantities and aspects of electromagnetism.

APPENDIX A: UNCERTAINTY ANALYSIS

The uncertainty in each length L_i and intensity I measurement is

$$\delta_{Li} = 1 \text{ mm}, \tag{7}$$
$$\delta_I = 0.01 \text{ units}. \tag{8}$$

Each length measurement for the mirror was performed independently twice per measurement and the measurements for d were made three times making the uncertainty in each measurement effectively

$$\delta_h = \frac{\delta_{Li}}{\sqrt{2}}, \tag{9}$$
$$\delta_d = \frac{\delta_{Li}}{\sqrt{3}} \tag{10}$$

The uncertainty in the path difference calculated is

$$\delta_{PD} = \sqrt{((d(\frac{d^2}{4}+h^2)^{\frac{-1}{2}}-1)\delta_d)^2 + (2h(\frac{d^2}{4}+h^2)^{\frac{-1}{2}}\delta_h)^2}$$

The slope m and y-intercept b of the linear least squares regression line between measured pairs of variables (x_i, y_i) are as follows:

$$m = \frac{\langle (x-\bar{x})(y-\bar{y})\rangle}{\langle (x-\bar{x})^2\rangle}, \tag{11}$$
$$b = \bar{y} - m\bar{x}. \tag{12}$$

These are determined by attempting to find a function that minimizes the squared difference between the measured quantities and the line of best fit.

The uncertainties in the slope and y-intercept are determined as follows:

$$\delta_m = \frac{\delta_y}{\sigma_x\sqrt{N}}, \tag{13}$$
$$\delta_b = \frac{\sigma_y}{\sqrt{N}}\sqrt{1+\left(\frac{\bar{x}}{\sigma_x}\right)^2}. \tag{14}$$

Where σ_x and σ_y is the statistical standard deviation of the population of x and y, and δ_y is the uncertainty in each measurement of y. It is important to note that these formulas assume that the uncertainties in each quantity are constant for each measurement.

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[1] P. K. Bondyopadhyay, IEEE Xplore **13** (2007).
[2] University of Connecticut Physics, Constructive and destructive interference, `https://www.phys.uconn.edu/~gibson/Notes/Section5_2/Sec5_2.htm` (2019), accessed: January 25, 2024.
[3] N. FOURIKIS, in *Advanced Array Systems, Applications and RF Technologies*, edited by N. FOURIKIS (Academic Press, 2000).